ENGG1300 Engineering Mechanics T3 2025

Faculty of Engineering

Shear Force and Bending Moment Lab Assignment

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Lab Timeslot:	**Monday 1-2pm**
Submission Date (dd/mm/yyyy):	**17/10/2025**

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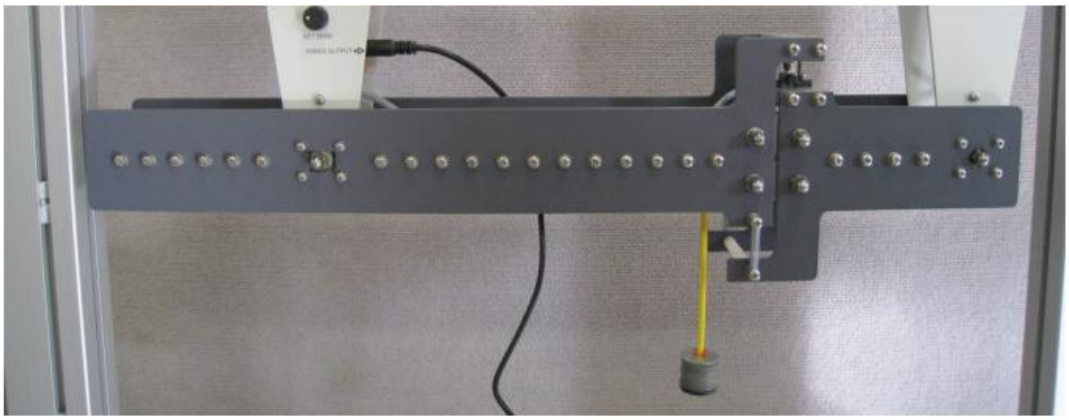
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Part A: Shear Force Experiment

Experiment 1: shear force variation with an increasing point load



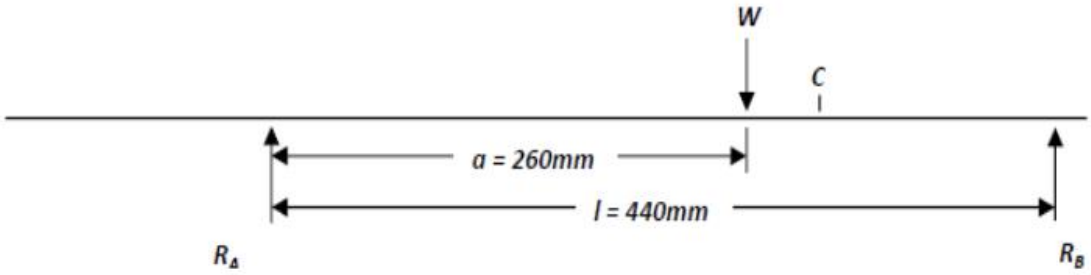


Figure 1 Experiment 1 set up and force diagram

This experiment examines how shear force varies with increasing point load. All students will perform the same experiment, using the same values for a, l and W, as shown in *Figure 1*.

What is the theoretical and experimental shear force at cut C? Use the following formula to compute the magnitude of the theoretical shear force

$$S_{c_{th}} = \frac{Wa}{l}$$

where a is the distance from the load, not the cut, to the left support. Note: This equation is only for experiment 1 and should not be used for the rest of the experiments. Remember, the experimental shear force at the cut in Newtons for all experiments is:

$$S_{c_{\exp}} = \text{displayed force.}$$

Mass (g)	Displayed force (N)	Load (N)	Theoretical shear force at the cut (N)	Experiment shear force at the cut (N)
100g	0.600	0.981	0.580	0.600
200g	1.200	1.962	1.160	1.200
300g	1.800	2.943	1.740	1.800
400g	2.400	3.924	2.319	2.400
500g	3.000	4.905	2.900	3.000

Place your typed and fully worked solution below:

To calculate the load weight, we can just get the mass and multiply it by the acceleration due to gravity

$$Load\ (Weight) = m \times g$$
$$= 0.1 \times 9.81$$
$$= 0.981(N).$$

To calculate the theoretical shear force, the formula provided above can be use as so,

$$S_{c_{th}} = \frac{Wa}{l}$$
$$= \frac{0.981 \times 0.260}{0.440}$$
$$= 0.580(N).$$

Experiment 2: shear force variation away from the point of loading

This experiment examines how shear force varies at the cut position for various loading conditions (W_1) and a depending on the student number, as shown in *Figure 2*.

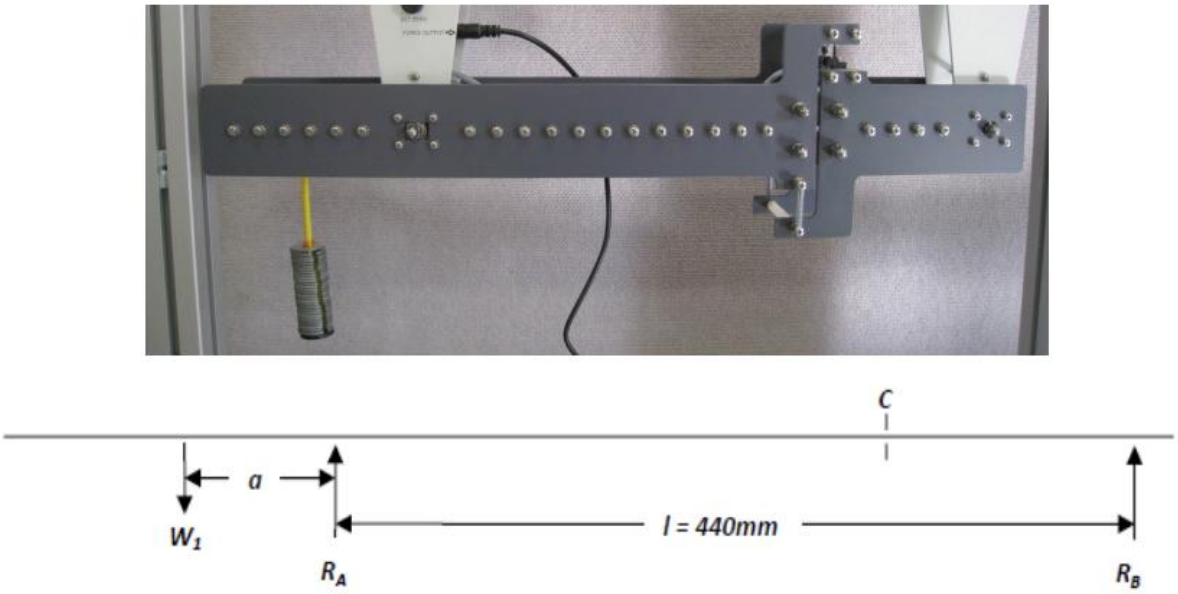


Figure 2 Experiment 2 set up and force diagram

The shear force at the cut position, C, is equal to the algebraic sum of the forces acting to the left and the right of C. Based on the force reading on the meter, calculate the support reactions R_A, R_B and calculate the theoretical shear force at the cut. Note: Depending on the sign convention chosen, the experimental and theoretical Shear Forces could have opposite signs. Therefore, you must specify your sign convention.

Force reading	0.5N upwards
Final value of R_A	$3.39N$ upwards
Final value of R_B	0.535N upwards
Theoretical shear force at the cut	0.535N downwards
Place your typed and fully worked solution below:	

$$m_1 = 0.4kg, a = 0.06m$$

The sum of forces int the y direction is

$$+\uparrow \sum F_y = 0$$
$$1: R_A + R_B - W_1 = 0.$$

The sum of moments about a is,

$$\sum M_A = 0$$

where positive is anticlockwise, so

$$2: R_B(l) - W_1(a) = 0,$$

and by rearranging we get

$$R_B = \frac{W_1(a)}{l}.$$

By rearranging equation 1 and 2 and solving for R_A we get

$$R_a = W_1 - \frac{W_1(a)}{l} = W_1\left(1 - \frac{a}{l}\right).$$

Using these equations the Final values can be calculated as so,

$$R_B = \frac{W_1(a)}{l} = \frac{0.400 \times 9.81 \times 0.060}{0.440} = 0.535(N)(upwards)$$

$$R_A = W_1 - R_B = 0.400 \times 9.81 - 0.535 = +3.39(N)(up).$$

Then by shear force conventions the theorical shear force is

$$V_{shear} = -R_B = -0.535(N)(downwards)$$

since if we take the right segment of the beam

$$+\uparrow \sum F_y = 0$$
$$R_b + V_{shear} = 0$$
$$V_{shear} = -R_b.$$

Although the signs differ from the experimental force reading the magnitude is similar, this could mean that the sign convention I have chosen differs from the machines. I have chosen positive shear to be upwards for the right face of the cut and hence positive to be downwards on the left side.

Experiment 3: shear force variation away from the point of loading

This experiment examines how shear force varies at the cut position for various loading conditions, as shown in *Figure 3*.

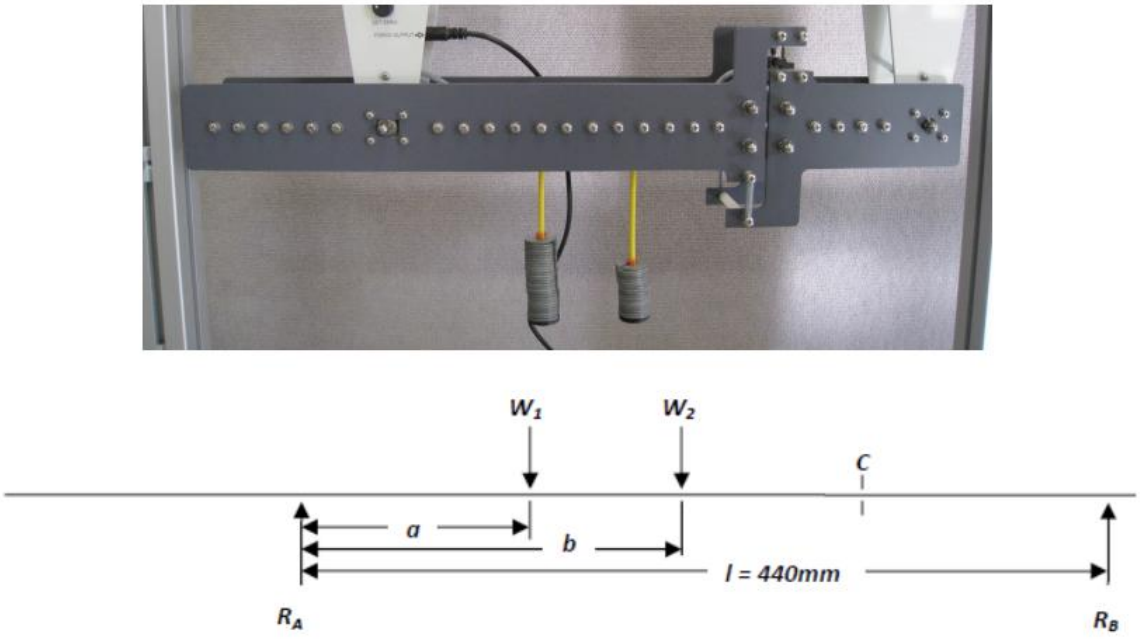


Figure 3 Experiment 3 set up and force diagram

The shear force at the cut position, C, is equal to the algebraic sum of the forces acting to the left and the right of C. Based on the force reading on the meter, calculate the support reactions R_A, R_B and the theoretical shear force at the cut. Note: Depending on the sign convention chosen, the experimental and theoretical Shear Forces could have opposite signs. Therefore, you must specify your sign convention.

Force reading	1.80N downwards
Final value of R_A	5.71N upwards
Final value of R_B	2.14N upwards
Theoretical shear force at the cut	2.14N upwards

Place your typed and fully worked solution below:

$$m_1 = 0.4kg, m_2 = 0.4kg, a = 0.06m, b = 0.18m$$

To calculate R_B we can just take the sum of moments around A

$$\sum M_A = 0$$
$$R_b(l) - W_1(a)R_b(l) - W_2(b) = 0$$

since the mass of both weights are the same

$$W_1 = W_2 = 0.4 \times 9.81 = 3.92(N)$$

$$R_b = \frac{3.924(0.06 + 0.180)}{0.44}$$
$$R_b = 2.14\ (N)\ upwards.$$

Then to calculate R_A we can use the sum of forces in the y directions

$$+\uparrow \sum F_y = 0$$
$$R_A + R_B - 2W_1 = 0$$

$$R_A = 5.71(N)\ upwards.$$

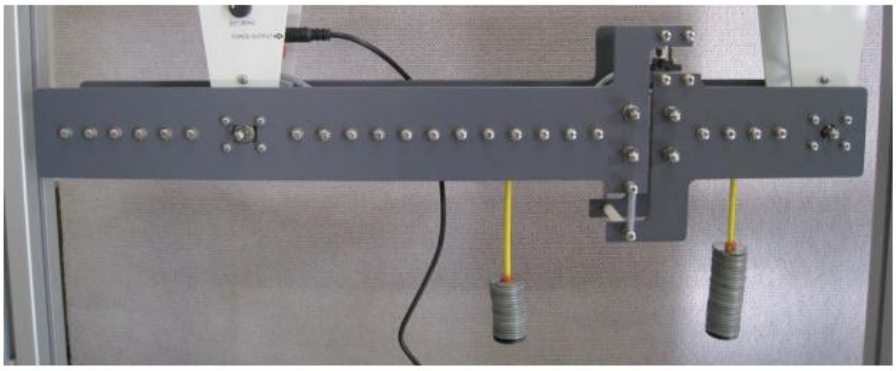
Finally finding the shear force is

$$R_A - W_1 - W_2 - V_{A_{shear}} = 0$$
$$V_{shear} = 5.71 - 2 \times 3.92$$
$$= -2.14(N)\ downwards.$$

Like the previous experiment the magnitude is similar, but the sign opposes each other which makes the aligns with the suspicion that different sign conventions is used.

Experiment 4: shear force variation away from the point of loading

This experiment examines how shear force varies at the cut position, for various loading conditions. Dimensions a, b and loads W_1, W_2 as shown in Figure 4 vary depending on the student number.



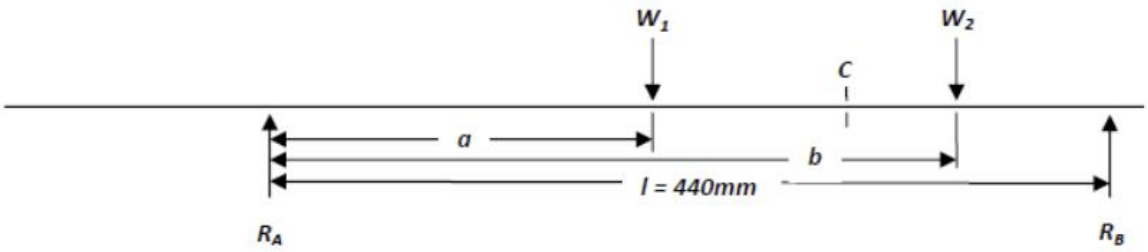


Figure 4 Experiment 4 set up and force diagram

The shear force at the cut position, C, is equal to the algebraic sum of the forces acting to the left and the right of C. Based on the force reading on the meter, calculate the support reactions R_A, R_B and the theoretical shear force at the cut. Note: Depending on the sign convention chosen, the experimental and theoretical Shear Forces could have opposite signs. Therefore, you must specify your sign convention.

Force reading	0.5N upwards
Final value of R_A	4.28N upwards
Final value of R_B	3.57N upwards

Theoretical shear force at the cut	0.357N downwards

Place your typed and fully worked solution below:

$$m_1 = 0.4kg, m_2 = 0.4kg, a = 0.06m, b = 0.34m$$

To calculate R_B we can just take the sum of moments around A

$$\sum M_A = 0$$
$$R_b(l) - W_1(a)R_b(l) - W_2(b) = 0$$

since the mass of both weights are the same

$$W_1 = W_2 = 0.4 \times 9.81 = 3.92(N)$$

$$R_b = \frac{3.924(0.06 + 0.340)}{0.44}$$
$$R_b = 3.567\ (N)\ upwards.$$

Then to calculate R_A we can use the sum of forces in the y directions

$$+\uparrow \sum F_y = 0$$
$$R_A + R_B - 2W_1 = 0$$

$$R_A = 4.28(N)\ upwards.$$

Finally finding the shear force is

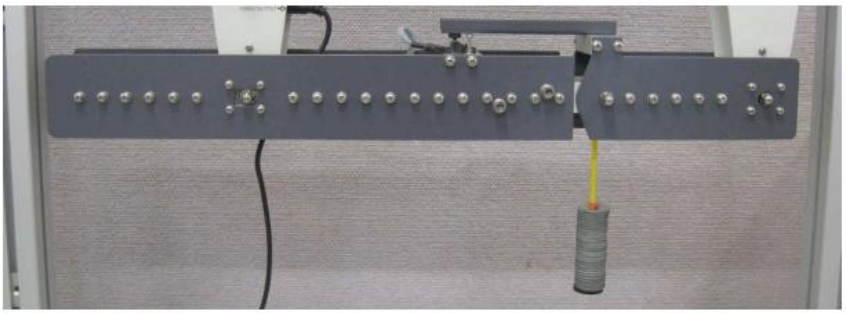
$$R_A - W_1 - V_{A_{shear}} = 0$$
$$V_{shear} = 4.28 - 3.92$$
$$= 0.357(N)\ downwards.$$

Sign is opposing again same reason as I stated previously.

Part B: Bending Moment in a Beam

Experiment 1: bending moment variation at the point of loading

This experiment examines how bending moment varies at the point of loading. All students will perform the same experiment, using the same values for a, l and W, as shown in *Figure 5*.



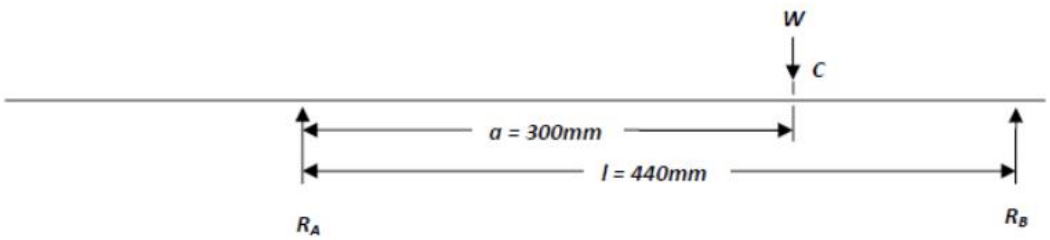


Figure 5 Experiment 1 set up and force diagram

Use the following equation to compute the theoretical bending moment at the cut

$$M_{C_{th}} = \frac{Wa(l-a)}{l},$$

and use the following equation to compute the experimental bending moment at the cut

$$M_{C_{\exp}} = \text{Displayed force} \times 0.125.$$

Mass (g)	Displayed force (N)	Load (N)	Theoretical bending moment at the cut (Nm)	Experiment bending moment at the cut (Nm)
100g	0.600	0.981	0.0936	0.075
200g	1.800	1.962	0.187	0.225
300g	2.300	2.943	0.281	0.288
400g	3.400	3.924	0.375	0.425
500g	4.300	4.905	0.468	0.538

Place your typed and fully worked solution below:

To calculate the load weight, we can just get the mass and multiply it by the acceleration due to gravity

$$\begin{aligned} Load\ (Weight) &= m \times g \\ &= 0.1 \times 9.81 \\ &= 0.981(N). \end{aligned}$$

To calculate the theoretical shear force, the formula provided above can be use as so,

$$\begin{aligned} M_{C_{th}} &= \frac{Wa(l-a)}{l} \\ &= \frac{0.981 \times 0.3(0.440 - 0.300)}{0.440} \\ &= 0.0936(N). \end{aligned}$$

To find the experimental value we can use the given equation like

$$\begin{aligned} M_{C_{\exp}} &= \text{Displayed force} \times 0.125 \\ &= 0.600 \times 0.125 \\ &= 0.075(N) \end{aligned}$$

Experiment 2: bending moment variation away from the point of loading

This experiment examines how bending moment varies at the cut position for various loading conditions (W_1) and a as shown in *Figure 6* depending on the student number.

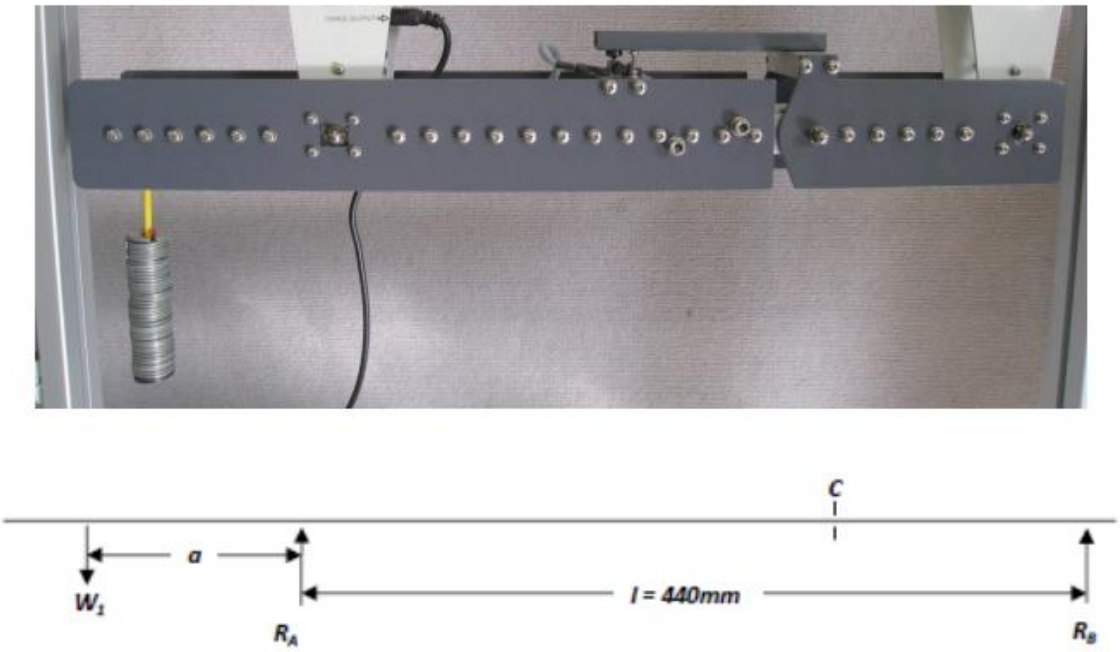


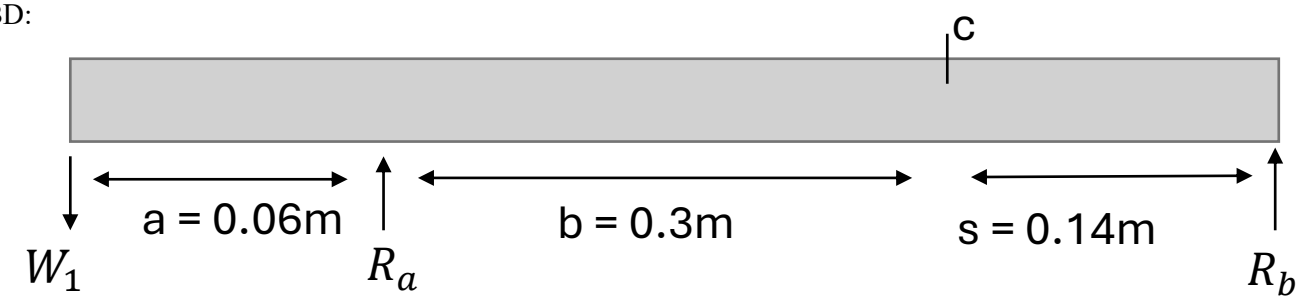
Figure 6 Experiment 2 set up and force diagram

The Bending Moment at the cut position C, is equal to the algebraic sum of the moments caused by the forces acting to the left and the right of C. Based on the force reading on the meter, determine the value of R_B for the calculation of the bending moment at C, since it will be easier to evaluate the bending moment with the single value of R_B than using W and R_A to the left of C.

Force reading	1.300N upwards
Final value of R_B	0.535N upwards
Bending moment at C	+0.0749Nm clockwise

Place your typed and fully worked solution below:

FBD:



$$m_1 = 0.4kg, a = 0.06m$$

Like experiment two in part A. the sum of moments about a is,

$$\sum M_A = 0$$

where positive is anticlockwise, so

$$R_B(l) - W_1(a) = 0,$$

and by rearranging we get

$$R_B = \frac{W_1(a)}{l}.$$

By rearranging equation 1 and 2 and solving for R_A we get

$$R_a = W_1 - \frac{W_1(a)}{l} = W_1\left(1 - \frac{a}{l}\right).$$

Using these equations the Final values can be calculated as so,

$$R_B = \frac{W_1(a)}{l} = \frac{0.400 \times 9.81 \times 0.060}{0.440} = 0.535(N)(upwards)$$

$$R_A = W_1 - R_B = 0.400 \times 9.81 - 0.535 = +3.39(N)(up).$$

Now to find the bending moment on the right side of c

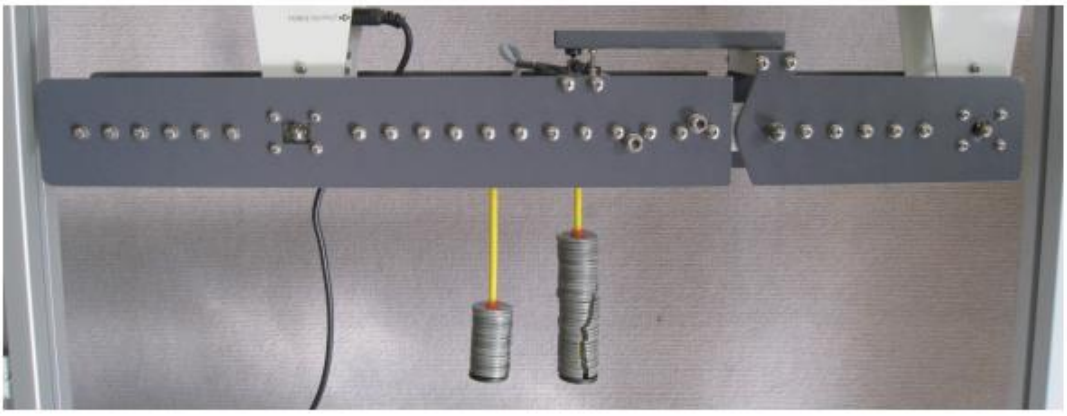
$$\begin{aligned}\sum M_c &= R_b \times \mathrm{s} \\ &= 0.535\mathrm{N} \times 0.14 \\ &= 0.0749(Nm).\end{aligned}$$

But because the this is the right side of the cut anticlockwise is negative and clockwise is positive, so the final answer is

$$+0.0749(Nm)\ clockwise$$

Experiment 3: bending moment variation away from the point of loading

This experiment examines how bending moment varies at the cut position for various loading conditions. Dimensions a, b and loads W_1, W_2 as shown in *Figure 7* vary depending on the student number.



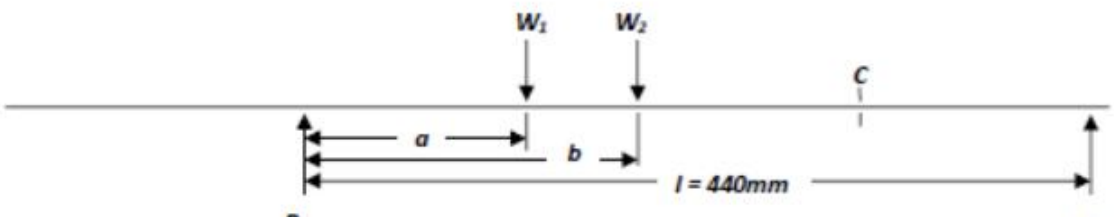
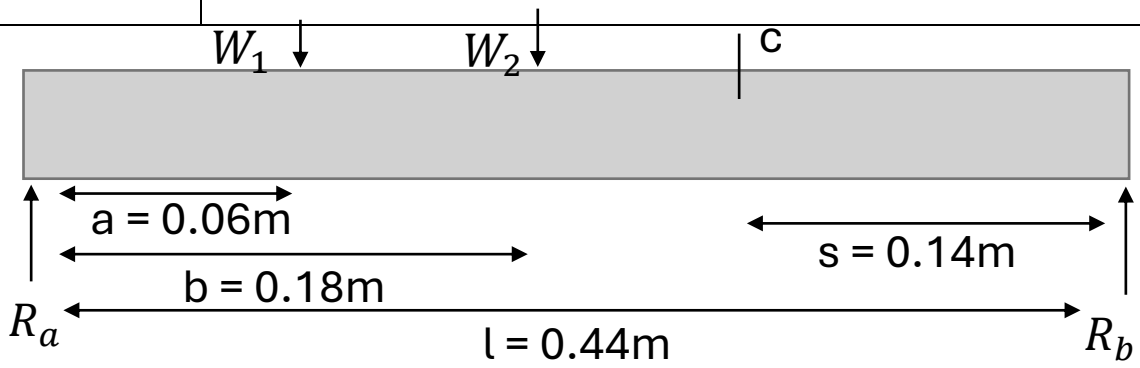


Figure 7 experiment 3 set up and force diagram

The Bending Moment at the cut position C, is equal to the algebraic sum of the moments caused by the forces acting to the left and the right of C. Based on the force reading, convert the force readings into bending moments (Nm). First calculate the support reactions R_A and R_B and then determine the bending moment at the cut C.

Force reading	2.50N downwards
Final value of R_A	5.71N upwards
Final value of R_B	2.14N upwards
Bending moment at C	+0.300Nm clockwise

FBD:



$$m_1 = 0.4kg, m_2 = 0.4kg, a = 0.06m, b = 0.18m$$

Similar to experiment three in part A, to calculate R_B we can just take the sum of moments around A

$$\sum M_A = 0$$
$$R_b(l) - W_1(a)R_b(l) - W_2(b) = 0$$

since the mass of both weights are the same

$$W_1 = W_2 = 0.4 \times 9.81 = 3.92(N)$$

$$R_b = \frac{3.924(0.06 + 0.180)}{0.44}$$
$$R_b = 2.14\ (N)\ upwards.$$

Then to calculate R_A we can use the sum of forces in the y directions

$$+\uparrow \sum F_y = 0$$
$$R_A + R_B - 2W_1 = 0$$

$$R_A = 5.71(N)\ upwards.$$

Now to find the bending moment at c

$$\sum M_c = R_b \times \text{s}$$
$$= 2.14\text{N} \times 0.14$$
$$= 0.2996(Nm).$$

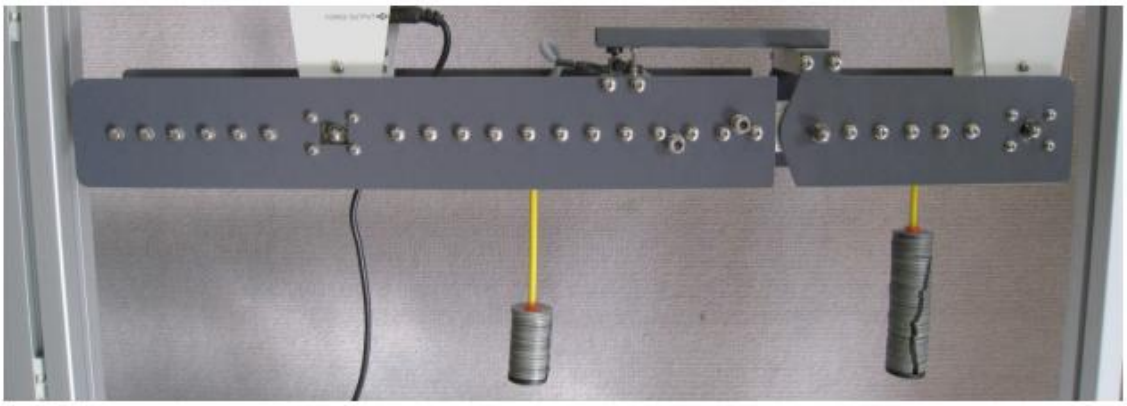
Like before, because the this is the right side of the cut anticlockwise is negative and clockwise is positive, so the final answer is

$$+0.2996(Nm)\ clockwise$$

Experiment 4: bending moment variation away from the point of loading

This experiment examines how bending moment varies at the cut position C for various loading conditions. s

The Bending Moment at the cut position C, is equal to the algebraic sum of the moments caused by the forces acting to



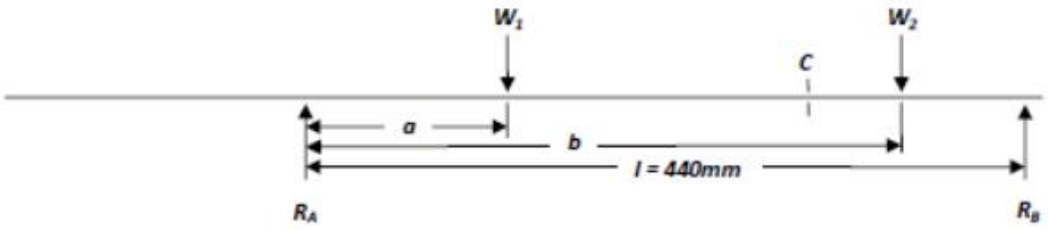


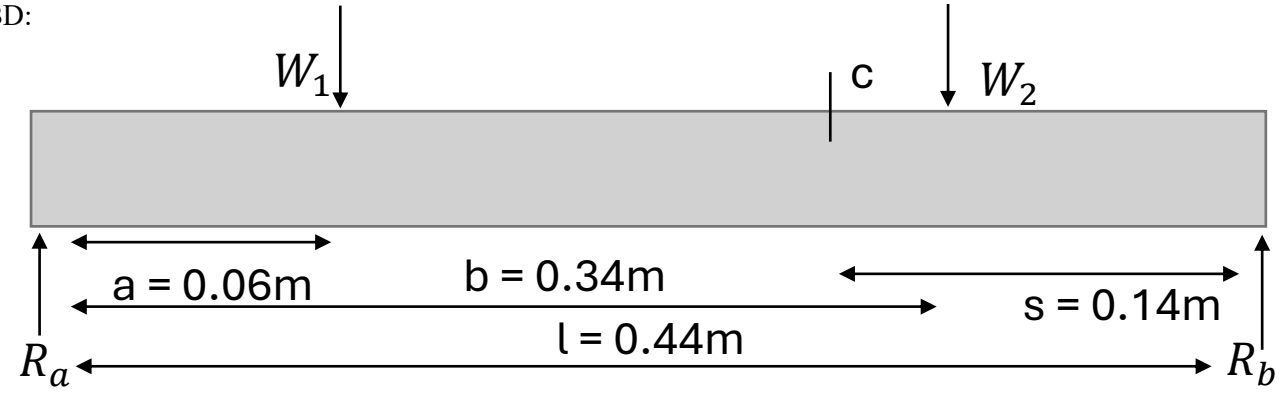
Figure 8 Experiment 4 setup and force diagram

the left and the right of C. Based on the force reading on the meter, convert the force readings into bending moments (Nm). First calculate the support reactions R_A and R_B and then determine the bending moment at the cut C.

Force reading	2.600N downwards
Final value of R_A	4.28N upwards
Final value of R_B	3.57N upwards
Bending moment at C	+0.343N clockwise

Place your typed and fully worked solution below:

FBD:



$$m_1 = 0.4kg, m_2 = 0.4kg, a = 0.06m, b = 0.34m$$

Like previously, to calculate R_B we can just take the sum of moments around A

$$\sum M_A = 0$$
$$R_b(l) - W_1(a)R_b(l) - W_2(b) = 0$$

since the mass of both weights are the same

$$W_1 = W_2 = 0.4 \times 9.81 = 3.92(N)$$

$$R_b = \frac{3.924(0.06 + 0.340)}{0.44}$$
$$R_b = 3.567\ (N)\ upwards.$$

Then to calculate R_A we can use the sum of forces in the y directions

$$+\uparrow \sum F_y = 0$$
$$R_A + R_B - 2W_1 = 0$$

$$R_A = 4.28(N)\ upwards.$$

Now to find the bending moment at c

$$\sum M_c = R_b \times \mathrm{s} - W_2 \times (\mathrm{s} - (\mathrm{l} - \mathrm{b}))$$
$$= 3.567\mathrm{N} \times 0.14 - 3.92 \times (0.14 - (0.44 - 0.34))$$
$$= 0.343(Nm).$$

Like before, because the this is the right side of the cut anticlockwise is negative and clockwise is positive, so the final answer is

$$+0.343(Nm)\ clockwise$$

Appendix A: Loading conditions and student number

The typical student number has 7 digits. Label these from the left as a, b, c, d, e, f & g (for example for the student number z3295406, $a = 3$ and $g = 6$). Loads and distances should be selected by each student by *Table 1*.

Table 1 Student number and the corresponding loads and geometry

z5686600

Load W_1 (for exp. 2, 3, 4)	**Load W_2 (for exp. 3, 4)**	**Distance a (for exp. 2, 3, 4)**	**Distance b (for exp. 3)**	**Distance b (for exp. 4)**
100g, if $d = 0, 1$	100g, if $e = 0, 1$	60mm, if $f = 0, 1$	180mm, if $g = 0, 1$	340mm, if $g = 0, 1$
200g, if $d = 2, 3$	200g, if $e = 2, 3$	80mm, if $f = 2, 3$	200mm, if $g = 2, 3$	340mm, if $g = 2, 3$
300g, if $d = 4, 5$	300g, if $e = 4, 5$	100mm, if $f = 4, 5$	220mm, if $g = 4, 5$	360mm, if $g = 4, 5$
400g, if $d = 6, 7$	400g, if $e = 6, 7$	120mm, if $f = 6, 7$	240mm, if $g = 6, 7$	380mm, if $g = 6, 7$
500g, if $d = 8, 9$	500g, if $e = 8, 9$	140mm, if $f = 8, 9$	260mm, if $g = 8, 9$	400mm, if $g = 8, 9$